Construction example

Instrument and available wood

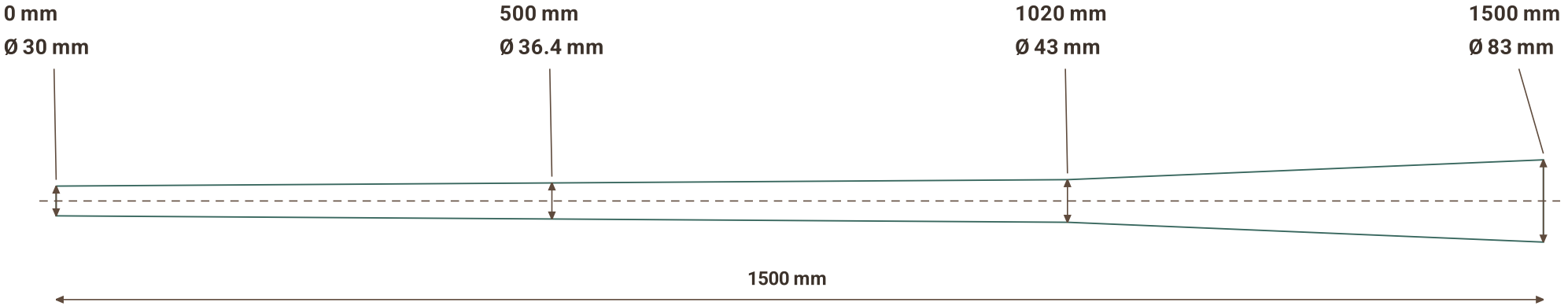
Instrument length: 1500 mm

Mouthpiece internal diameter: 30 mm

Bell internal diameter: 83 mm

Target note: C♯2 · 69.3 Hz

Available wood



White: final air passage · Longitudinal view · scale 1:6

Wood dimensions have not been supplied

Not measured yet

Minimum per half, before preparation: width 101 mm × thickness 50.5 mm

Minimum wall thickness: 8 mm

Extra carving allowance: 1 mm

Wood measurements still need confirmation

More measurements are needed to check the full length

Wood condition and grain still need assessment. The target note is not a guarantee of the played note.

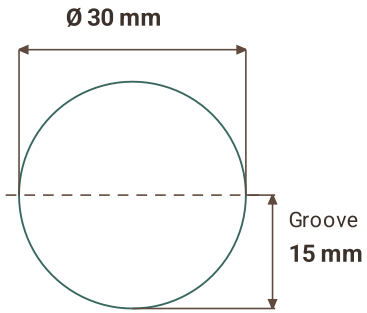
100 mm

100 mm - print at 100% and verify with a ruler

Construction example

Interior and carving dimensions

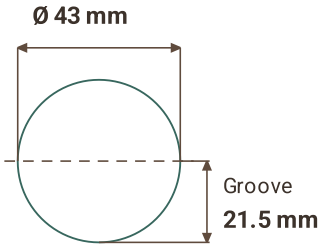
0 mm



Not measured yet

Cross-section scale 1:1

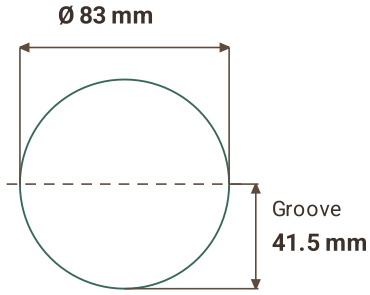
1020 mm



Not measured yet

Cross-section scale 1:2

1500 mm



Not measured yet

Cross-section scale 1:3

Carving measurements · millimetres

From the mouthpiece	Internal bore diameter	Groove depth per half	Remaining wall thickness
0	30	15	Not confirmed
200	32.5	16.3	Not confirmed
500	36.4	18.2	Not confirmed
750	39.6	19.8	Not confirmed
1020	43	21.5	Not confirmed
1300	66.3	33.2	Not confirmed
1500	83	41.5	Not confirmed

Selected stations; complete measurements in CSV

Depth is measured from the joining plane; remaining wood is below the groove. Dimensions allow for the stated preparation losses.



100 mm - print at 100% and verify with a ruler